AMD User Guide

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Abstract

AMD is a set of routines that implements the approximate minimum degree ordering algorithm to permute sparse matrices prior to numerical factorization. There are versions written in both C and Fortran 77. A MATLAB interface is included.

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Availability: http://www.suitesparse.com.

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1 Overview

AMD is a set of routines for preordering a sparse matrix prior to numerical factorization. It uses an approximate minimum degree ordering algorithm [1, 2] to find a permutation matrix $\mathbf{P}$ so that the Cholesky factorization $\mathbf{PAP}^{\mathsf{T}} = \mathbf{LL}^{\mathsf{T}}$ has fewer (often much fewer) nonzero entries than the Cholesky factorization of $\mathbf{A}$. The algorithm is typically much faster than other ordering methods and minimum degree ordering algorithms that compute an exact degree [4]. Some methods, such as approximate deficiency [9] and graph-partitioning based methods [5, 7, 8, 10] can produce better orderings, depending on the matrix.

The algorithm starts with an undirected graph representation of a symmetric sparse matrix $\mathbf{A}$. Node i in the graph corresponds to row and column i of the matrix, and there is an edge (i, j) in the graph if a_{ij} is nonzero. The degree of a node is initialized to the number of off-diagonal nonzeros in row i, which is the size of the set of nodes adjacent to i in the graph.

The selection of a pivot a_{ii} from the diagonal of $\mathbf{A}$ and the first step of Gaussian elimination corresponds to one step of graph elimination. Numerical fill-in causes new nonzero entries in the matrix (fill-in refers to nonzeros in $\mathbf{L}$ that are not in $\mathbf{A}$). Node i is eliminated and edges are added to its neighbors so that they form a clique (or *element*). To reduce fill-in, node i is selected as the node of least degree in the graph. This process repeats until the graph is eliminated.

The clique is represented implicitly. Rather than listing all the new edges in the graph, a single list of nodes is kept which represents the clique. This list corresponds to the nonzero pattern of the first column of $\mathbf{L}$. As the elimination proceeds, some of these cliques become subsets of subsequent cliques, and are removed. This graph can be stored in place, that is using the same amount of memory as the original graph.

The most costly part of the minimum degree algorithm is the recomputation of the degrees of nodes adjacent to the current pivot element. Rather than keep track of the exact degree, the approximate minimum degree algorithm finds an upper bound on the degree that is easier to compute. For nodes of least degree, this bound tends to be tight. Using the approximate degree instead of the exact degree leads to a substantial savings in run time, particularly for very irregularly structured matrices. It has no effect on the quality of the ordering.

In the C version of AMD, the elimination phase is followed by an elimination tree post-ordering. This has no effect on fill-in, but reorganizes the ordering so that the subsequent numerical factorization is more efficient. It also includes a pre-processing phase in which nodes of very high degree are removed (without causing fill-in), and placed last in the permutation $\mathbf{P}$. This reduces the run time substantially if the matrix has a few rows with many nonzero entries, and has little effect on the quality of the ordering. The C version operates on the symmetric nonzero pattern of $\mathbf{A}+\mathbf{A}^{\mathsf{T}}$, so it can be given an unsymmetric matrix, or either the lower or upper triangular part of a symmetric matrix.

The two Fortran versions of AMD are essentially identical to two versions of the AMD algorithm discussed in an earlier paper [1] (approximate minimum external degree, both with and without aggressive absorption). For a discussion of the long history of the minimum degree algorithm, see [4].

2 Availability

In addition to appearing as a Collected Algorithm of the ACM,
AMD is available at http://www.suitesparse.com. The Fortran version is available as the routine `MC47` in HSL (formerly the Harwell Subroutine Library) [6].

3 Using AMD in MATLAB

The MATLAB function `amd` is now a built-in function in MATLAB 7.3 (R2006b). The built-in `amd` and the `amd2` function provided here differ in how the optional parameters are passed (the 2nd input parameter).

To use AMD2 in MATLAB, you must first compile the AMD2 mexFunction. Just type `make` in the Unix system shell, while in the `AMD/MATLAB` directory. You can also type `amd_make` in MATLAB, while in the `AMD/MATLAB` directory. Place the `AMD/MATLAB` directory in your MATLAB path. This can be done on any system with MATLAB, including Windows. See Section 8 for more details on how to install AMD.

The MATLAB statement `p=amd(A)` finds a permutation vector `p` such that the Cholesky factorization `chol(A(p,p))` is typically sparser than `chol(A)`. If `A` is unsymmetric, `amd(A)` is identical to `amd(A+A')` (ignoring numerical cancellation). If `A` is not symmetric positive definite, but has substantial diagonal entries and a mostly symmetric nonzero pattern, then this ordering is also suitable for LU factorization. A partial pivoting threshold may be required to prevent pivots from being selected off the diagonal, such as the statement `[L,U,P] = lu (A (p,p), 0.1)`. Type `help lu` for more details. The statement `[L,U,P,Q] = lu (A (p,p))` in MATLAB 6.5 is not suitable, however, because it uses UMFPACK Version 4.0 and thus does not attempt to select pivots from the diagonal. UMFPACK Version 4.1 in MATLAB 7.0 and later uses several strategies, including a symmetric pivoting strategy, and will give you better results if you want to factorize an unsymmetric matrix of this type. Refer to the UMFPACK User Guide for more details, at http://www.suitesparse.com.

The AMD mexFunction is much faster than the built-in MATLAB symmetric minimum degree ordering methods, SYMAMD and SYMMMD. Its ordering quality is comparable to SYMAMD, and better than SYMMMD [3].

An optional input argument can be used to modify the control parameters for AMD (aggressive absorption, dense row/column handling, and printing of statistics). An optional output argument provides statistics on the ordering, including an analysis of the fill-in and the floating-point operation count for a subsequent factorization. For more details (once AMD is installed), type `help amd` in the MATLAB command window.

4 Using AMD in a C program

The C-callable AMD library consists of eight user-callable routines and one include file. There are two versions of seven of the routines, with `int32_t` and `int64_t` integers. The routines with prefix `amd_l_` use `int64_t` integer arguments; the others use `int32_t` integer arguments.

The following routines are fully described in Section 9:

- `amd_order` (`int64_t` version: `amd_l_order`)

```
#include "amd.h"
int32_t n, Ap [n+1], Ai [nz], P [n] ;
double Control [AMD_CONTROL], Info [AMD_INFO] ;
int result = amd_order (n, Ap, Ai, P, Control, Info) ;
```

  Computes the approximate minimum degree ordering of an n-by-n matrix $\mathbf{A}$. Returns a permutation vector `P` of size `n`, where `P[k] = i` if row and column `i` are the `k`th row and column in the permuted matrix. This routine allocates its own memory of size $1.2e + 9n$ integers, where e is the number of nonzeros in $\mathbf{A} + \mathbf{A}^{\mathsf{T}}$. It computes statistics about the

matrix **A**, such as the symmetry of its nonzero pattern, the number of nonzeros in **L**, and the number of floating-point operations required for Cholesky and LU factorizations (which are returned in the `Info` array). The user's input matrix is not modified. It returns `AMD_OK` if successful, `AMD_OK_BUT_JUMBLED` if successful (but the matrix had unsorted and/or duplicate row indices), `AMD_INVALID` if the matrix is invalid, `AMD_OUT_OF_MEMORY` if out of memory.

- `amd_defaults` (`int64_t` version: `amd_l_defaults`)

```
#include "amd.h"
double Control [AMD_CONTROL] ;
amd_defaults (Control) ;
```

  Sets the default control parameters in the `Control` array. These can then be modified as desired before passing the array to the other AMD routines.

- `amd_control` (`int64_t` version: `amd_l_control`)

```
#include "amd.h"
double Control [AMD_CONTROL] ;
amd_control (Control) ;
```

  Prints a description of the control parameters, and their values.

- `amd_info` (`int64_t` version: `amd_l_info`)

```
#include "amd.h"
double Info [AMD_INFO] ;
amd_info (Info) ;
```

  Prints a description of the statistics computed by AMD, and their values.

- `amd_valid` (`int64_t` version: `amd_l_valid`)

```
#include "amd.h"
int32_t n, Ap [n+1], Ai [nz] ;
int result = amd_valid (n, n, Ap, Ai) ;
```

  Returns `AMD_OK` or `AMD_OK_BUT_JUMBLED` if the matrix is valid as input to `amd_order`; the latter is returned if the matrix has unsorted and/or duplicate row indices in one or more columns. Returns `AMD_INVALID` if the matrix cannot be passed to `amd_order`. For `amd_order`, the matrix must also be square. The first two arguments are the number of rows and the number of columns of the matrix. For its use in AMD, these must both equal `n`.

- `amd_2` (`int64_t` version: `amd_l2`) AMD ordering kernel. It is faster than `amd_order`, and can be called by the user, but it is difficult to use. It does not check its inputs for errors. It does not require the columns of its input matrix to be sorted, but it destroys the matrix on output. Additional workspace must be passed. Refer to the source file `AMD/Source/amd_2.c` for a description.

- `amd_version`: returns the AMD version.

The nonzero pattern of the matrix $\mathbf{A}$ is represented in compressed column form. For an n-by-n matrix $\mathbf{A}$ with `nz` nonzero entries, the representation consists of two arrays: `Ap` of size `n+1` and `Ai` of size `nz`. The row indices of entries in column `j` are stored in `Ai[Ap[j] ... Ap[j+1]-1]`. For `amd_order`, if duplicate row indices are present, or if the row indices in any given column are not sorted in ascending order, then `amd_order` creates an internal copy of the matrix with sorted rows and no duplicate entries, and orders the copy. This adds slightly to the time and memory usage of `amd_order`, but is not an error condition.

The matrix is 0-based, and thus row indices must be in the range `0` to `n-1`. The first entry `Ap[0]` must be zero. The total number of entries in the matrix is thus `nz = Ap[n]`.

The matrix must be square, but it does not need to be symmetric. The `amd_order` routine constructs the nonzero pattern of $\mathbf{B} = \mathbf{A} + \mathbf{A}^{\mathsf{T}}$ (without forming $\mathbf{A}^{\mathsf{T}}$ explicitly if $\mathbf{A}$ has sorted columns and no duplicate entries), and then orders the matrix $\mathbf{B}$. Thus, either the lower triangular part of $\mathbf{A}$, the upper triangular part, or any combination may be passed. The transpose $\mathbf{A}^{\mathsf{T}}$ may also be passed to `amd_order`. The diagonal entries may be present, but are ignored.

4.1 Control parameters

Control parameters are set in an optional `Control` array. It is optional in the sense that if a `NULL` pointer is passed for the `Control` input argument, then default control parameters are used.

- `Control[AMD_DENSE]` (or `Control(1)` in MATLAB): controls the threshold for "dense" rows/columns. A dense row/column in $\mathbf{A} + \mathbf{A}^{\mathsf{T}}$ can cause AMD to spend significant time in ordering the matrix. If `Control[AMD_DENSE]` ≥ 0, rows/columns with more than `Control[AMD_DENSE]` $\sqrt{n}$ entries are ignored during the ordering, and placed last in the output order. The default value of `Control[AMD_DENSE]` is 10. If negative, no rows/columns are treated as "dense." Rows/columns with 16 or fewer off-diagonal entries are never considered "dense."

- `Control[AMD_AGGRESSIVE]` (or `Control(2)` in MATLAB): controls whether or not to use aggressive absorption, in which a prior element is absorbed into the current element if it is a subset of the current element, even if it is not adjacent to the current pivot element (refer to [1, 2] for more details). The default value is nonzero, which means that aggressive absorption will be performed. This nearly always leads to a better ordering (because the approximate degrees are more accurate) and a lower execution time. There are cases where it can lead to a slightly worse ordering, however. To turn it off, set `Control[AMD_AGGRESSIVE]` to 0.

Statistics are returned in the `Info` array (if `Info` is `NULL`, then no statistics are returned). Refer to `amd.h` file, for more details (14 different statistics are returned, so the list is not included here).

4.2 Sample C program

The following program, `amd_demo.c`, illustrates the basic use of AMD. See Section 5 for a short description of each calling sequence.

```
#include "amd.h"

int32_t n = 5 ;
int32_t Ap [ ] = { 0,   2,       6,       10,  12, 14} ;
int32_t Ai [ ] = { 0,1, 0,1,2,4, 1,2,3,4, 2,3, 1,4   } ;
int32_t P [5] ;

int main (void)
```

```
{
    int32_t k ;
    (void) amd_order (n, Ap, Ai, P, (double *) NULL, (double *) NULL) ;
    for (k = 0 ; k < n ; k++) printf ("P [%d] = %d\n", k, P [k]) ;
    return (0) ;
}
```

The `Ap` and `Ai` arrays represent the binary matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

The diagonal entries are ignored. AMD constructs the pattern of $\mathbf{A}+\mathbf{A}^{\mathsf{T}}$, and returns a permutation vector of $(0, 3, 1, 4, 2)$. Since the matrix is unsymmetric but with a mostly symmetric nonzero pattern, this would be a suitable permutation for an LU factorization of a matrix with this nonzero pattern and whose diagonal entries are not too small. The program uses default control settings and does not return any statistics about the ordering, factorization, or solution (`Control` and `Info` are both `(double *) NULL`). It also ignores the status value returned by `amd_order`.

More example programs are included with the AMD package. The `amd_demo.c` program provides a more detailed demo of AMD. Another example is the AMD mexFunction, `amd_mex.c`.

4.3 A note about zero-sized arrays

AMD uses several user-provided arrays of size `n` or `nz`. Either `n` or `nz` can be zero. If you attempt to `malloc` an array of size zero, however, `malloc` will return a null pointer which AMD will report as invalid. If you `malloc` an array of size `n` or `nz` to pass to AMD, make sure that you handle the `n` = 0 and `nz = 0` cases correctly.

5 Synopsis of C-callable routines

The matrix **A** is `n`-by-`n` with `nz` entries.

```
#include "amd.h"
int32_t n, status, Ap [n+1], Ai [nz], P [n] ;
double Control [AMD_CONTROL], Info [AMD_INFO] ;
amd_defaults (Control) ;
status = amd_order (n, Ap, Ai, P, Control, Info) ;
amd_control (Control) ;
amd_info (Info) ;
status = amd_valid (n, n, Ap, Ai) ;
```

The `amd_l_*` routines are identical, except that all `int32_t` arguments become `int64_t`:

```
#include "amd.h"
int64_t n, status, Ap [n+1], Ai [nz], P [n] ;
double Control [AMD_CONTROL], Info [AMD_INFO] ;
amd_l_defaults (Control) ;
status = amd_l_order (n, Ap, Ai, P, Control, Info) ;
amd_l_control (Control) ;
amd_l_info (Info) ;
status = amd_l_valid (n, n, Ap, Ai) ;
```

The amd_version function uses plain int:

```
#include "amd.h"
int version [3] ;
amd_version (version) ;
```

6 Using AMD in a Fortran program

Two Fortran versions of AMD are provided. The AMD routine computes the approximate minimum degree ordering, using aggressive absorption. The AMDBAR routine is identical, except that it does not perform aggressive absorption. The AMD routine is essentially identical to the HSL routine MC47B/BD. Note that earlier versions of the Fortran AMD and AMDBAR routines included an IOVFLO argument, which is no longer present.

In contrast to the C version, the Fortran routines require a symmetric nonzero pattern, with no diagonal entries present although the MC47A/AD wrapper in HSL allows duplicates, ignores out-of-range entries, and only uses entries from the upper triangular part of the matrix. Although we have an experimental Fortran code for treating "dense" rows, the Fortran codes in this release do not treat "dense" rows and columns of **A** differently, and thus their run time can be high if there are a few dense rows and columns in the matrix. They do not perform a post-ordering of the elimination tree, compute statistics on the ordering, or check the validity of their input arguments. These facilities are provided by MC47A/AD and other subroutines from HSL. Only one integer version of each Fortran routine is provided. Both Fortran routines overwrite the user's input matrix, in contrast to the C version. The C version does not return the elimination or assembly tree. The Fortran version returns an assembly tree; refer to the User Guide for details. The following is the syntax of the AMD Fortran routine. The AMDBAR routine is identical except for the routine name.

```
      INTEGER N, IWLEN, PFREE, NCMPA, IW (IWLEN), PE (N), DEGREE (N), NV (N),
     $         NEXT (N), LAST (N), HEAD (N), ELEN (N), W (N), LEN (N)
      CALL AMD (N, PE, IW, LEN, IWLEN, PFREE, NV, NEXT,
     $         LAST, HEAD, ELEN, DEGREE, NCMPA, W)
      CALL AMDBAR (N, PE, IW, LEN, IWLEN, PFREE, NV, NEXT,
     $         LAST, HEAD, ELEN, DEGREE, NCMPA, W)
```

The input matrix is provided to AMD and AMDBAR in three arrays, PE, of size N, LEN, of size N, and IW, of size IWLEN. The size of IW must be at least NZ+N. The recommended size is 1.2*NZ + N. On input, the indices of nonzero entries in row I are stored in IW. PE(I) is the index in IW of the start of row I. LEN(I) is the number of entries in row I. The matrix is 1-based, with row and column indices in the range 1 to N. Row I is contained in IW (PE(I) ... PE(I) + LEN(I) - 1). The diagonal entries must not be present. The indices within each row must not contain any duplicates, but they need not be sorted. The rows themselves need not be in any particular order, and there may be empty space between the rows. If LEN(I) is zero, then there are no off-diagonal entries in row I, and PE(I) is ignored. The integer PFREE defines what part of IW contains the user's input matrix, which is held in IW(1 ... PFREE-1). The contents of IW and LEN are undefined on output, and PE is modified to contain information about the ordering.

As the algorithm proceeds, it modifies the IW array, placing the pattern of the partially eliminated matrix in IW(PFREE ... IWLEN). If this space is exhausted, the space is compressed. The number of compressions performed on the IW array is returned in the scalar NCMPA. The value of PFREE on output is the length of IW required for no compressions to be needed.

The output permutation is returned in the array LAST, of size N. If I=LAST(K), then I is the Kth row in the permuted matrix. The inverse permutation is returned in the array ELEN, where

K=ELEN(I) if I is the Kth row in the permuted matrix. On output, the PE and NV arrays hold the assembly tree, a supernodal elimination tree that represents the relationship between columns of the Cholesky factor **L**. If NV(I) > 0, then I is a node in the assembly tree, and the parent of I is -PE(I). If I is a root of the tree, then PE(I) is zero. The value of NV(I) is the number of entries in the corresponding column of **L**, including the diagonal. If NV(I) is zero, then I is a non-principal node that is not in the assembly tree. Node -PE(I) is the parent of node I in a subtree, the root of which is a node in the assembly tree. All nodes in one subtree belong to the same supernode in the assembly tree. The other size N arrays (DEGREE, HEAD, NEXT, and W) are used as workspace, and are not defined on input or output.

If you want to use a simpler user-interface and compute the elimination tree post-ordering, you should be able to call the C routines amd_order or amd_l_order from a Fortran program. Just be sure to take into account the 0-based indexing in the P, Ap, and Ai arguments to amd_order and amd_l_order. A sample interface is provided in the files AMD/Demo/amd_f77cross.f and AMD/Demo/amd_f77wrapper.c. To compile the amd_f77cross program, type make cross in the AMD/Demo directory. The Fortran-to-C calling conventions are highly non-portable, so this example is not guaranteed to work with your compiler C and Fortran compilers. The output of amd_f77cross is in amd_f77cross.out.

7 Sample Fortran main program

The following program illustrates the basic usage of the Fortran version of AMD. The AP and AI arrays represent the binary matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

in a conventional 1-based column-oriented form, except that the diagonal entries are not present. The matrix has the same as nonzero pattern of $\mathbf{A} + \mathbf{A}^{\mathsf{T}}$ in the C program, in Section 4. The output permutation is $(4, 1, 3, 5, 2)$. It differs from the permutation returned by the C routine amd_order because a post-order of the elimination tree has not yet been performed.

```
      INTEGER N, NZ, J, K, P, IWLEN, PFREE, NCMPA
      PARAMETER (N = 5, NZ = 10, IWLEN = 17)
      INTEGER AP (N+1), AI (NZ), LAST (N), PE (N), LEN (N), ELEN (N),
     $     IW (IWLEN), DEGREE (N), NV (N), NEXT (N), HEAD (N), W (N)
      DATA AP / 1, 2,     5,     8,  9,  11/
      DATA AI / 2, 1,3,5, 2,4,5, 3,  2,3   /
C     load the matrix into the AMD workspace
      DO 10 J = 1,N
          PE (J) = AP (J)
          LEN (J) = AP (J+1) - AP (J)
10    CONTINUE
      DO 20 P = 1,NZ
          IW (P) = AI (P)
20    CONTINUE
      PFREE = NZ + 1
C     order the matrix (destroys the copy of A in IW, PE, and LEN)
      CALL AMD (N, PE, IW, LEN, IWLEN, PFREE, NV, NEXT, LAST, HEAD,
     $     ELEN, DEGREE, NCMPA, W)
```

```
      DO 60 K = 1, N
         PRINT 50, K, LAST (K)
50       FORMAT ('P (',I2,') = ', I2)
60    CONTINUE
      END
```

The `Demo` directory contains an example of how the C version may be called from a Fortran program, but this is highly non-portable. For this reason, it is placed in the `Demo` directory, not in the primary `Source` directory.

8 Installation

AMD now relies primarily on CMake to build the library. It also includes a simple `AMD/Makefile` which uses cmake to do the actual build. The use of this `AMD/Makefile` is optional; for Windows, just import the CMakeLists.txt into MS Visual Studio.

To compile and install the library for both system-wide usage and local usage:

```
make
sudo make install
```

To compile/install for just local usage (SuiteSparse/lib and SuiteSparse/include):

```
make local
make install
```

To run the demos

```
make demos
```

To remove all files in `AMD/` not in the original distribution (leaves SuiteSparse/lib and SuiteSparse/include unchanged):

```
make clean
```

To use the AMD2 mexFunction in MATLAB, simply type `amd_make` in MATLAB while in the `AMD/MATLAB` directory. This works on any system with MATLAB, including Windows. Alternatively, just use the built-in `amd` in MATLAB 7.3 or later, which is the same as this package.

9 The AMD routines

The file `AMD/Include/amd.h` listed below describes each user-callable routine in the C version of AMD, and gives details on their use.

```
//------------------------------------------------------------------------------
// AMD/Include/amd.h:  approximate minimum degree ordering
//------------------------------------------------------------------------------

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// Iain S. Duff.  All Rights Reserved.
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//------------------------------------------------------------------------------

/* AMD finds a symmetric ordering P of a matrix A so that the Cholesky
 * factorization of P*A*P' has fewer nonzeros and takes less work than the
 * Cholesky factorization of A.  If A is not symmetric, then it performs its
 * ordering on the matrix A+A'.  Two sets of user-callable routines are
 * provided, one for int32_t integers and the other for int64_t integers.
 *
 * The method is based on the approximate minimum degree algorithm, discussed
 * in Amestoy, Davis, and Duff, "An approximate degree ordering algorithm",
 * SIAM Journal of Matrix Analysis and Applications, vol. 17, no. 4, pp.
 * 886-905, 1996.  This package can perform both the AMD ordering (with
 * aggressive absorption), and the AMDBAR ordering (without aggressive
 * absorption) discussed in the above paper.  This package differs from the
 * Fortran codes discussed in the paper:
 *
 *       (1) it can ignore "dense" rows and columns, leading to faster run times
 *       (2) it computes the ordering of A+A' if A is not symmetric
 *       (3) it is followed by a depth-first post-ordering of the assembly tree
 *           (or supernodal elimination tree)
 *
 * For historical reasons, the Fortran versions, amd.f and amdbar.f, have
 * been left (nearly) unchanged.  They compute the identical ordering as
 * described in the above paper.
 */

#ifndef AMD_H
#define AMD_H

#include "SuiteSparse_config.h"

/* make it easy for C++ programs to include AMD */
#ifdef __cplusplus
extern "C" {
#endif

int amd_order  /* returns AMD_OK, AMD_OK_BUT_JUMBLED,
                                  * AMD_INVALID, or AMD_OUT_OF_MEMORY */
(
    int32_t n,                     /* A is n-by-n.  n must be >= 0. */
    const int32_t Ap [ ],          /* column pointers for A, of size n+1 */
    const int32_t Ai [ ],          /* row indices of A, of size nz = Ap [n] */
    int32_t P [ ],                 /* output permutation, of size n */
    double Control [ ],     /* input Control settings, of size AMD_CONTROL */
    double Info [ ]         /* output Info statistics, of size AMD_INFO */
) ;
```

```
int amd_l_order  /* see above for description */
(
    int64_t n,
    const int64_t Ap [ ],
    const int64_t Ai [ ],
    int64_t P [ ],
    double Control [ ],
    double Info [ ]
) ;

/* Input arguments (not modified):
 *
 *      n: the matrix A is n-by-n.
 *      Ap: an int32_t/int64_t array of size n+1, containing column
 *          pointers of A.
 *      Ai: an int32_t/int64_t array of size nz, containing the row
 *          indices of A, where nz = Ap [n].
 *      Control:  a double array of size AMD_CONTROL, containing control
 *          parameters.  Defaults are used if Control is NULL.
 *
 * Output arguments (not defined on input):
 *
 *      P: an int32_t/int64_t array of size n, containing the output
 *          permutation. If row i is the kth pivot row, then P [k] = i.  In
 *          MATLAB notation, the reordered matrix is A (P,P).
 *      Info: a double array of size AMD_INFO, containing statistical
 *          information.  Ignored if Info is NULL.
 *
 * On input, the matrix A is stored in column-oriented form.  The row indices
 * of nonzero entries in column j are stored in Ai [Ap [j] ... Ap [j+1]-1].
 *
 * If the row indices appear in ascending order in each column, and there
 * are no duplicate entries, then amd_order is slightly more efficient in
 * terms of time and memory usage.  If this condition does not hold, a copy
 * of the matrix is created (where these conditions do hold), and the copy is
 * ordered.
 *
 * Row indices must be in the range 0 to
 * n-1.  Ap [0] must be zero, and thus nz = Ap [n] is the number of nonzeros
 * in A.  The array Ap is of size n+1, and the array Ai is of size nz = Ap [n].
 * The matrix does not need to be symmetric, and the diagonal does not need to
 * be present (if diagonal entries are present, they are ignored except for
 * the output statistic Info [AMD_NZDIAG]).  The arrays Ai and Ap are not
 * modified.  This form of the Ap and Ai arrays to represent the nonzero
 * pattern of the matrix A is the same as that used internally by MATLAB.
 * If you wish to use a more flexible input structure, please see the
 * umfpack_*_triplet_to_col routines in the UMFPACK package, at
 * http://www.suitesparse.com.
 *
 * Restrictions:  n >= 0.  Ap [0] = 0.  Ap [j] <= Ap [j+1] for all j in the
 *      range 0 to n-1.  nz = Ap [n] >= 0.  Ai [0..nz-1] must be in the range 0
 *      to n-1.  Finally, Ai, Ap, and P must not be NULL.  If any of these
 *      restrictions are not met, AMD returns AMD_INVALID.
 *
 * AMD returns:
 *
 *      AMD_OK if the matrix is valid and sufficient memory can be allocated to
 *          perform the ordering.
```

```
*
*      AMD_OUT_OF_MEMORY if not enough memory can be allocated.
*
*      AMD_INVALID if the input arguments n, Ap, Ai are invalid, or if P is
*          NULL.
*
*      AMD_OK_BUT_JUMBLED if the matrix had unsorted columns, and/or duplicate
*          entries, but was otherwise valid.
*
* The AMD routine first forms the pattern of the matrix A+A', and then
* computes a fill-reducing ordering, P.  If P [k] = i, then row/column i of
* the original is the kth pivotal row.  In MATLAB notation, the permuted
* matrix is A (P,P), except that 0-based indexing is used instead of the
* 1-based indexing in MATLAB.
*
* The Control array is used to set various parameters for AMD.  If a NULL
* pointer is passed, default values are used.  The Control array is not
* modified.
*
*      Control [AMD_DENSE]:  controls the threshold for "dense" rows/columns.
*          A dense row/column in A+A' can cause AMD to spend a lot of time in
*          ordering the matrix.  If Control [AMD_DENSE] >= 0, rows/columns
*          with more than Control [AMD_DENSE] * sqrt (n) entries are ignored
*          during the ordering, and placed last in the output order.  The
*          default value of Control [AMD_DENSE] is 10.  If negative, no
*          rows/columns are treated as "dense".  Rows/columns with 16 or
*          fewer off-diagonal entries are never considered "dense".
*
*      Control [AMD_AGGRESSIVE]: controls whether or not to use aggressive
*          absorption, in which a prior element is absorbed into the current
*          element if is a subset of the current element, even if it is not
*          adjacent to the current pivot element (refer to Amestoy, Davis,
*          & Duff, 1996, for more details).  The default value is nonzero,
*          which means to perform aggressive absorption.  This nearly always
*          leads to a better ordering (because the approximate degrees are
*          more accurate) and a lower execution time.  There are cases where
*          it can lead to a slightly worse ordering, however.  To turn it off,
*          set Control [AMD_AGGRESSIVE] to 0.
*
*      Control [2..4] are not used in the current version, but may be used in
*          future versions.
*
* The Info array provides statistics about the ordering on output.  If it is
* not present, the statistics are not returned.  This is not an error
* condition.
*
*      Info [AMD_STATUS]:  the return value of AMD, either AMD_OK,
*          AMD_OK_BUT_JUMBLED, AMD_OUT_OF_MEMORY, or AMD_INVALID.
*
*      Info [AMD_N]: n, the size of the input matrix
*
*      Info [AMD_NZ]: the number of nonzeros in A, nz = Ap [n]
*
*      Info [AMD_SYMMETRY]:  the symmetry of the matrix A.  It is the number
*          of "matched" off-diagonal entries divided by the total number of
*          off-diagonal entries.  An entry A(i,j) is matched if A(j,i) is also
*          an entry, for any pair (i,j) for which i != j.  In MATLAB notation,
*              S = spones (A) ;
*              B = tril (S, -1) + triu (S, 1) ;
```

```
 *              symmetry = nnz (B & B') / nnz (B) ;
 *
 *      Info [AMD_NZDIAG]: the number of entries on the diagonal of A.
 *
 *      Info [AMD_NZ_A_PLUS_AT]:  the number of nonzeros in A+A', excluding the
 *          diagonal.  If A is perfectly symmetric (Info [AMD_SYMMETRY] = 1)
 *          with a fully nonzero diagonal, then Info [AMD_NZ_A_PLUS_AT] = nz-n
 *          (the smallest possible value).  If A is perfectly unsymmetric
 *          (Info [AMD_SYMMETRY] = 0, for an upper triangular matrix, for
 *          example) with no diagonal, then Info [AMD_NZ_A_PLUS_AT] = 2*nz
 *          (the largest possible value).
 *
 *      Info [AMD_NDENSE]: the number of "dense" rows/columns of A+A' that were
 *          removed from A prior to ordering.  These are placed last in the
 *          output order P.
 *
 *      Info [AMD_MEMORY]: the amount of memory used by AMD, in bytes.  In the
 *          current version, this is 1.2 * Info  [AMD_NZ_A_PLUS_AT] + 9*n
 *          times the size of an integer.  This is at most 2.4nz + 9n.  This
 *          excludes the size of the input arguments Ai, Ap, and P, which have
 *          a total size of nz + 2*n + 1 integers.
 *
 *      Info [AMD_NCMPA]: the number of garbage collections performed.
 *
 *      Info [AMD_LNZ]: the number of nonzeros in L (excluding the diagonal).
 *          This is a slight upper bound because mass elimination is combined
 *          with the approximate degree update.  It is a rough upper bound if
 *          there are many "dense" rows/columns.  The rest of the statistics,
 *          below, are also slight or rough upper bounds, for the same reasons.
 *          The post-ordering of the assembly tree might also not exactly
 *          correspond to a true elimination tree postordering.
 *
 *      Info [AMD_NDIV]: the number of divide operations for a subsequent LDL'
 *          or LU factorization of the permuted matrix A (P,P).
 *
 *      Info [AMD_NMULTSUBS_LDL]:  the number of multiply-subtract pairs for a
 *          subsequent LDL' factorization of A (P,P).
 *
 *      Info [AMD_NMULTSUBS_LU]:  the number of multiply-subtract pairs for a
 *          subsequent LU factorization of A (P,P), assuming that no numerical
 *          pivoting is required.
 *
 *      Info [AMD_DMAX]:  the maximum number of nonzeros in any column of L,
 *          including the diagonal.
 *
 *      Info [14..19] are not used in the current version, but may be used in
 *          future versions.
 */

/* ---------------------------------------------------------------------- */
/* direct interface to AMD */
/* ---------------------------------------------------------------------- */

/* amd_2 is the primary AMD ordering routine.  It is not meant to be
 * user-callable because of its restrictive inputs and because it destroys
 * the user's input matrix.  It does not check its inputs for errors, either.
 * However, if you can work with these restrictions it can be faster than
 * amd_order and use less memory (assuming that you can create your own copy
 * of the matrix for AMD to destroy).  Refer to AMD/Source/amd_2.c for a
```

```
 * description of each parameter. */

void amd_2
(
    int32_t n,
    int32_t Pe [ ],
    int32_t Iw [ ],
    int32_t Len [ ],
    int32_t iwlen,
    int32_t pfree,
    int32_t Nv [ ],
    int32_t Next [ ],
    int32_t Last [ ],
    int32_t Head [ ],
    int32_t Elen [ ],
    int32_t Degree [ ],
    int32_t W [ ],
    double Control [ ],
    double Info [ ]
) ;

void amd_l2
(
    int64_t n,
    int64_t Pe [ ],
    int64_t Iw [ ],
    int64_t Len [ ],
    int64_t iwlen,
    int64_t pfree,
    int64_t Nv [ ],
    int64_t Next [ ],
    int64_t Last [ ],
    int64_t Head [ ],
    int64_t Elen [ ],
    int64_t Degree [ ],
    int64_t W [ ],
    double Control [ ],
    double Info [ ]
) ;

/* ------------------------------------------------------------------------ */
/* amd_valid */
/* ------------------------------------------------------------------------ */

/* Returns AMD_OK or AMD_OK_BUT_JUMBLED if the matrix is valid as input to
 * amd_order; the latter is returned if the matrix has unsorted and/or
 * duplicate row indices in one or more columns.  Returns AMD_INVALID if the
 * matrix cannot be passed to amd_order.  For amd_order, the matrix must also
 * be square.  The first two arguments are the number of rows and the number
 * of columns of the matrix.  For its use in AMD, these must both equal n.
 */

int amd_valid
(
    int32_t n_row,                /* # of rows */
    int32_t n_col,                /* # of columns */
    const int32_t Ap [ ],         /* column pointers, of size n_col+1 */
    const int32_t Ai [ ]          /* row indices, of size Ap [n_col] */
) ;
```

```
int amd_l_valid
(
    int64_t n_row,
    int64_t n_col,
    const int64_t Ap [ ],
    const int64_t Ai [ ]
) ;

/* ---------------------------------------------------------------------- */
/* AMD Control and Info arrays */
/* ---------------------------------------------------------------------- */

/* amd_defaults:  sets the default control settings */
void amd_defaults   (double Control [ ]) ;
void amd_l_defaults (double Control [ ]) ;

/* amd_control: prints the control settings */
void amd_control    (double Control [ ]) ;
void amd_l_control  (double Control [ ]) ;

/* amd_info: prints the statistics */
void amd_info       (double Info [ ]) ;
void amd_l_info     (double Info [ ]) ;

// amd_version: return AMD version.  The version array is returned with
// version [0..2] = {AMD_MAIN_VERSION, AMD_SUB_VERSION, AMD_SUBSUB_VERSION}
void amd_version (int version [3]) ;

#ifdef __cplusplus
}
#endif

#define AMD_CONTROL 5          /* size of Control array */
#define AMD_INFO 20            /* size of Info array */

/* contents of Control */
#define AMD_DENSE 0            /* "dense" if degree > Control [0] * sqrt (n) */
#define AMD_AGGRESSIVE 1    /* do aggressive absorption if Control [1] != 0 */

/* default Control settings */
#define AMD_DEFAULT_DENSE 10.0          /* default "dense" degree 10*sqrt(n) */
#define AMD_DEFAULT_AGGRESSIVE 1    /* do aggressive absorption by default */

/* contents of Info */
#define AMD_STATUS 0           /* return value of amd_order and amd_l_order */
#define AMD_N 1                /* A is n-by-n */
#define AMD_NZ 2      /* number of nonzeros in A */
#define AMD_SYMMETRY 3         /* symmetry of pattern (1 is sym., 0 is unsym.) */
#define AMD_NZDIAG 4           /* # of entries on diagonal */
#define AMD_NZ_A_PLUS_AT 5  /* nz in A+A' */
#define AMD_NDENSE 6           /* number of "dense" rows/columns in A */
#define AMD_MEMORY 7           /* amount of memory used by AMD */
#define AMD_NCMPA 8            /* number of garbage collections in AMD */
#define AMD_LNZ 9     /* approx. nz in L, excluding the diagonal */
#define AMD_NDIV 10            /* number of fl. point divides for LU and LDL' */
#define AMD_NMULTSUBS_LDL 11 /* number of fl. point (*,-) pairs for LDL' */
#define AMD_NMULTSUBS_LU 12  /* number of fl. point (*,-) pairs for LU */
#define AMD_DMAX 13             /* max nz. in any column of L, incl. diagonal */
```

```
/* ------------------------------------------------------------------------ */
/* return values of AMD */
/* ------------------------------------------------------------------------ */

#define AMD_OK 0           /* success */
#define AMD_OUT_OF_MEMORY -1        /* malloc failed, or problem too large */
#define AMD_INVALID -2              /* input arguments are not valid */
#define AMD_OK_BUT_JUMBLED 1        /* input matrix is OK for amd_order, but
    * columns were not sorted, and/or duplicate entries were present.  AMD had
    * to do extra work before ordering the matrix.  This is a warning, not an
    * error.  */

/* ========================================================================== */
/* === AMD version ========================================================== */
/* ========================================================================== */

/* AMD Version 1.2 and later include the following definitions.
 * As an example, to test if the version you are using is 1.2 or later:
 *
 * #ifdef AMD_VERSION
 *       if (AMD_VERSION >= AMD_VERSION_CODE (1,2)) ...
 * #endif
 *
 * This also works during compile-time:
 *
 *       #if defined(AMD_VERSION) && (AMD_VERSION >= AMD_VERSION_CODE (1,2))
 *           printf ("This is version 1.2 or later\n") ;
 *       #else
 *           printf ("This is an early version\n") ;
 *       #endif
 *
 * Versions 1.1 and earlier of AMD do not include a #define'd version number.
 */

#define AMD_DATE "June 1, 2025"
#define AMD_MAIN_VERSION   3
#define AMD_SUB_VERSION    3
#define AMD_SUBSUB_VERSION 4

#define AMD_VERSION_CODE(main,sub) SUITESPARSE_VER_CODE(main,sub)
#define AMD_VERSION AMD_VERSION_CODE(3,3)

#define AMD__VERSION SUITESPARSE__VERCODE(3,3,4)
#if !defined (SUITESPARSE__VERSION) || \
    (SUITESPARSE__VERSION < SUITESPARSE__VERCODE(7,11,0))
#error "AMD 3.3.4 requires SuiteSparse_config 7.11.0 or later"
#endif

#endif
```

References

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